SCHEDULE 1

Team: EXA2

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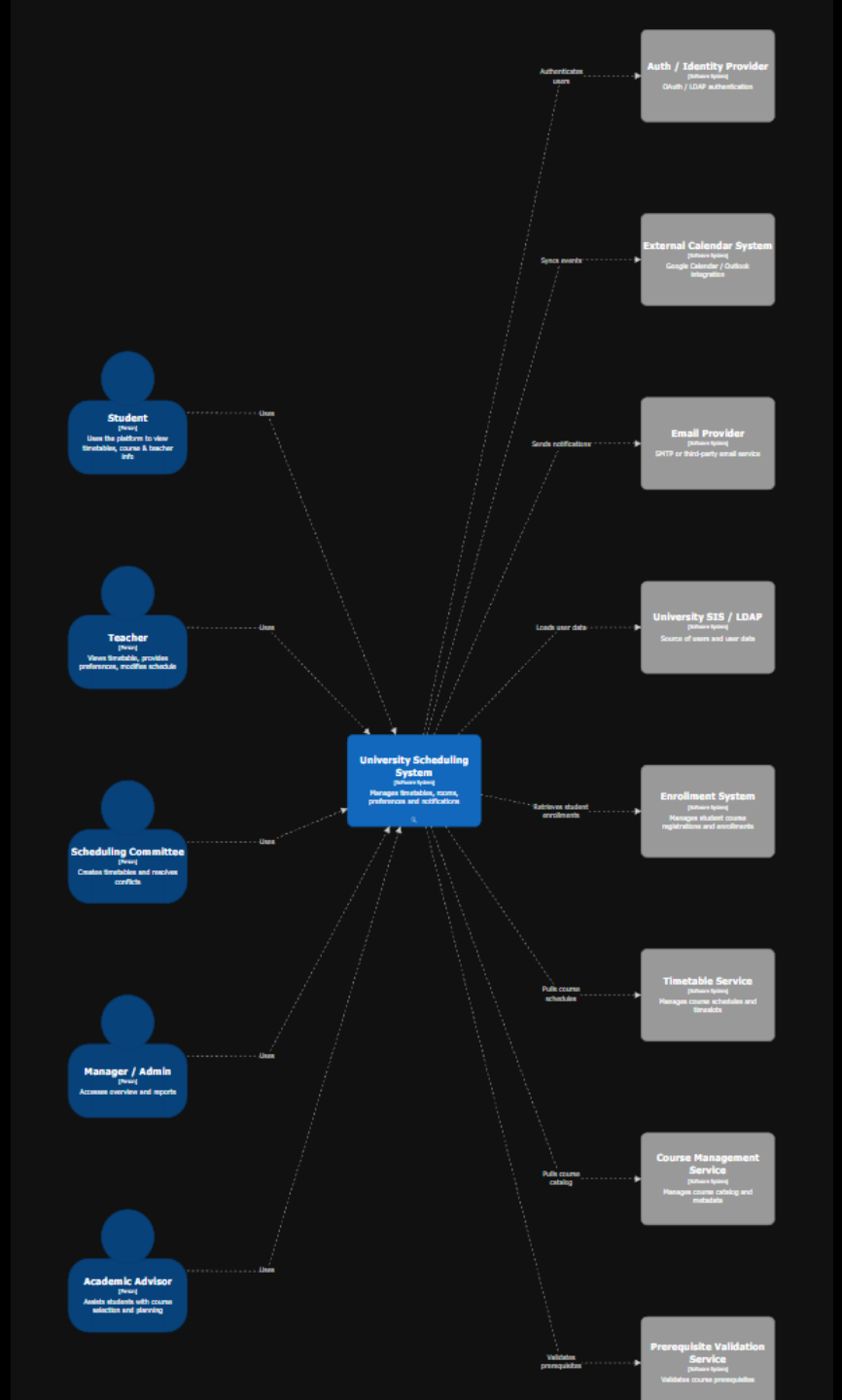
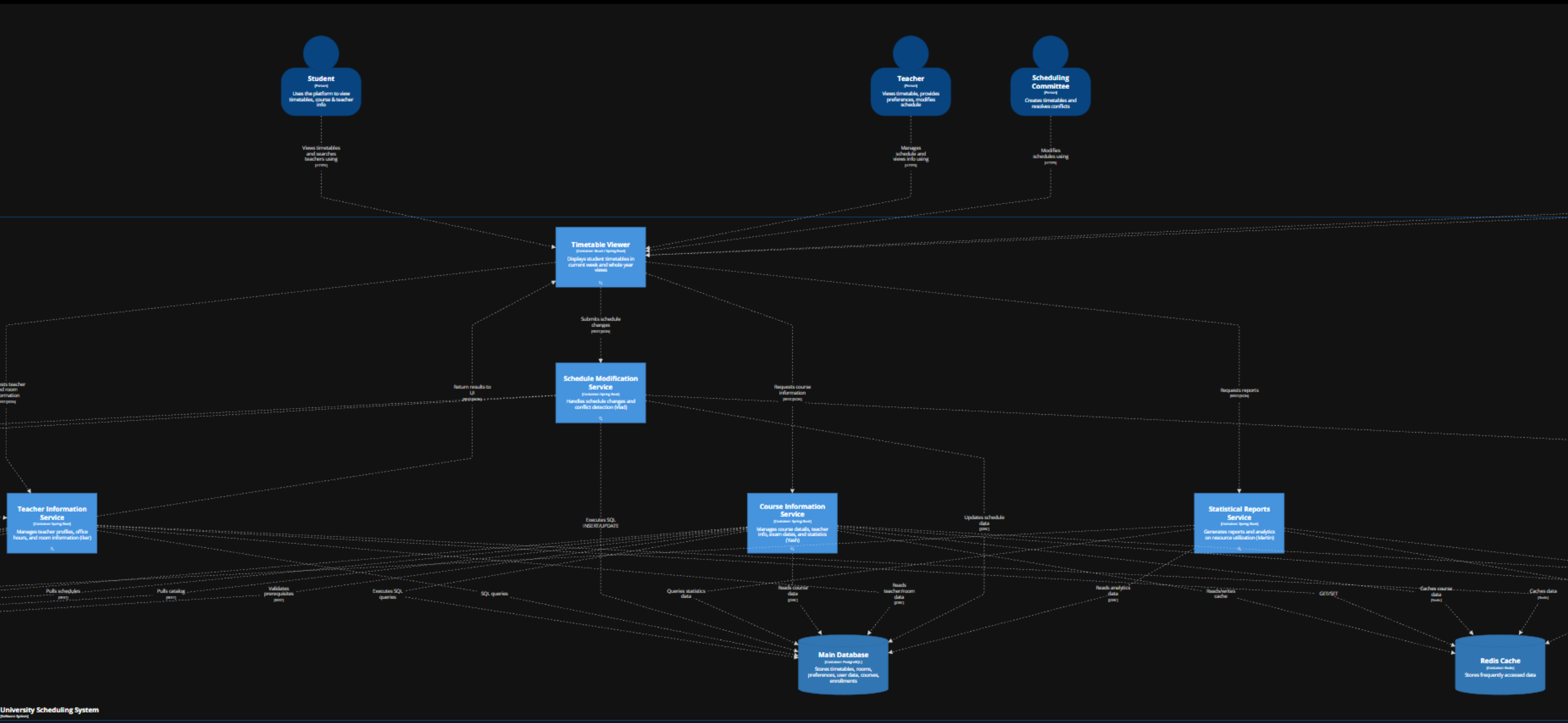
University wants to changes the login system

- Source of Stimulus: Developer
- Stimulus: University changes the Auth/Identity Provider
- Artifact: System-level auth integration and containers/components that enforce permissions, e.g.: scheduleModificationContainer.authorizationService, role validation inside teacherInfoContainer (Auth/Role Validator), and the system’s connection to authProvider.
- Environment: Design-time.
- Response: Developers update auth integration (configuration/adapter + role mapping), update affected services, then test and deploy the change to staging and production.
- Measure: Change implemented + deployed in ≤ 3 developer-days

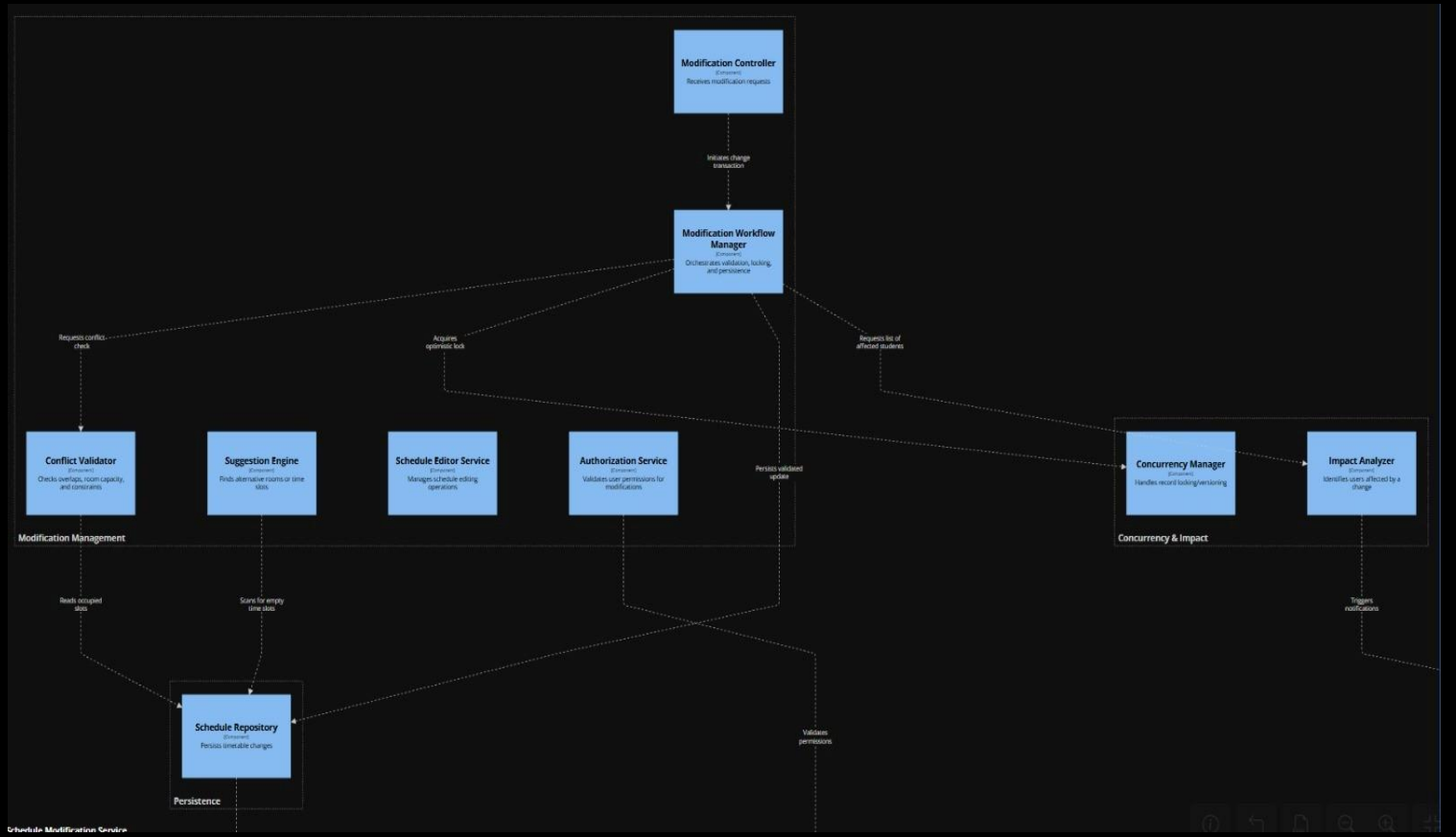

Student
[Person]
Uses the platform to view timetables, course & teacher info
Teacher
[Person]
Views timetable, provides preferences, modifies schedule
Scheduling Committee
[Person]
Creates timetables and resolves conflicts
Manager / Admin
[Person]
Accesses overview and reports
Academic Advisor
[Person]
Assists students with course selection and planning
Uses
University Scheduling System
[Software System]
Manages timetables, rooms, preferences and notifications
Authenticates users
Auth / Identity Provider
[Software System]
OAuth / LDAP authentication
Syncs events
External Calendar System
[Software System]
Google Calendar / Outlook integration
Sends notifications
Email Provider
[Software System]
SMTP or third-party email service
Loads user data
University SIS / LDAP
[Software System]
Source of users and user data
Retrieves student enrollments
Enrollment System
[Software System]
Manages student course registrations and enrollments
Pulls course schedules
Timetable Service
[Software System]
Manages course schedules and timeslots
Pulls course catalog
Course Management Service
[Software System]
Manages course catalog and metadata
Validates prerequisites
Prerequisite Validation Service
[Software System]
Validates course prerequisites


Student
[Person]
Uses the platform to view timetables, course & teacher info
Teacher
[Person]
Views timetable, provides preferences, modifies schedule
Scheduling Committee
[Person]
Creates timetables and resolves conflicts
Views timetables and searches teachers using [HTTPS]
Manages schedule and views info using [HTTPS]
Modifies schedules using [HTTPS]
Timetable Viewer
[Container: React / Spring Boot]
Displays student timetables in current week and whole year views
Submits schedule changes [HTTP/JSON]
Schedule Modification Service
[Container: Spring Boot]
Handles schedule changes and conflict detection (Vlad)
Return results to UI [HTTP/JSON]
Requests course information [HTTP/JSON]
Requests reports [HTTP/JSON]
Teacher Information Service
[Container: Spring Boot]
Manages teacher profiles, office hours, and room information (Ikar)
Course Information Service
[Container: Spring Boot]
Manages course details, teacher info, exam dates, and statistics (Vasih)
Statistical Reports Service
[Container: Spring Boot]
Generates reports and analytics on resource utilization (Martin)
Executes SQL INSERT/UPDATE
Updates schedule data [JDBC]
Pulls schedules [REST]
Pulls catalog [REST]
Validates prerequisites [REST]
Executes SQL queries
SQL queries
Queries statistics data
Reads course data [JDBC]
Reads teacher/room data [JDBC]
Reads analytics data [JDBC]
Reads/writes cache
GET/SET
Caches course data [Redis]
Caches data [Redis]
Main Database
[Container: PostgreSQL]
Stores timetables, rooms, preferences, user data, courses, enrollments
Redis Cache
[Container: Redis]
Stores frequently accessed data
University Scheduling System
[Software System]

Emil Farzaliyev

Concurrency conflict during Timetable Modification

- Source of Stimulus: Scheduling Committee member
- Stimulus: Two committee members try to modify the same course timeslot/room at nearly the same time (concurrent update).
- Artifact: Schedule Modification Service - specifically Modification Workflow Manager + Concurrency Manager + Schedule Repository
- Environment: Production, during timetable-finalization period.
- Response: The system saves the first change. When the second person tries to save, the system notices the schedule was already changed by someone else, so it does not overwrite it. Instead, it shows a “someone updated this” message and loads the newest schedule so the second person can try again. The event is logged.
- Measure: 100% of conflicting concurrent edits are detected and rejectedConflict response returned within ≤ 2 seconds



Modifiability – Ease of Feature Extension (Schedule Export)

- Source of Stimulus: University administration / product owner
- Stimulus: Add a feature to export student schedules to Google Calendar / iCal.

- Artifact:
  - timetableViewer (Timetable Viewer)
  - loadBalancer (Nginx / AWS ALB)
  - calendarIntegrationService (new)
  - scheduleModificationContainer (Schedule Modification Service)
  - externalCalendar (External Calendar System)

- Environment: Production, normal operation.

- Response: The Timetable Viewer sends an export request via the Load Balancer. The Calendar Integration Service fetches schedule data from the Schedule Modification Service (REST/JSON) and creates/updates events in the External Calendar System.

- Measure: Feature added by introducing a new service; existing services remain unchanged (no regression).

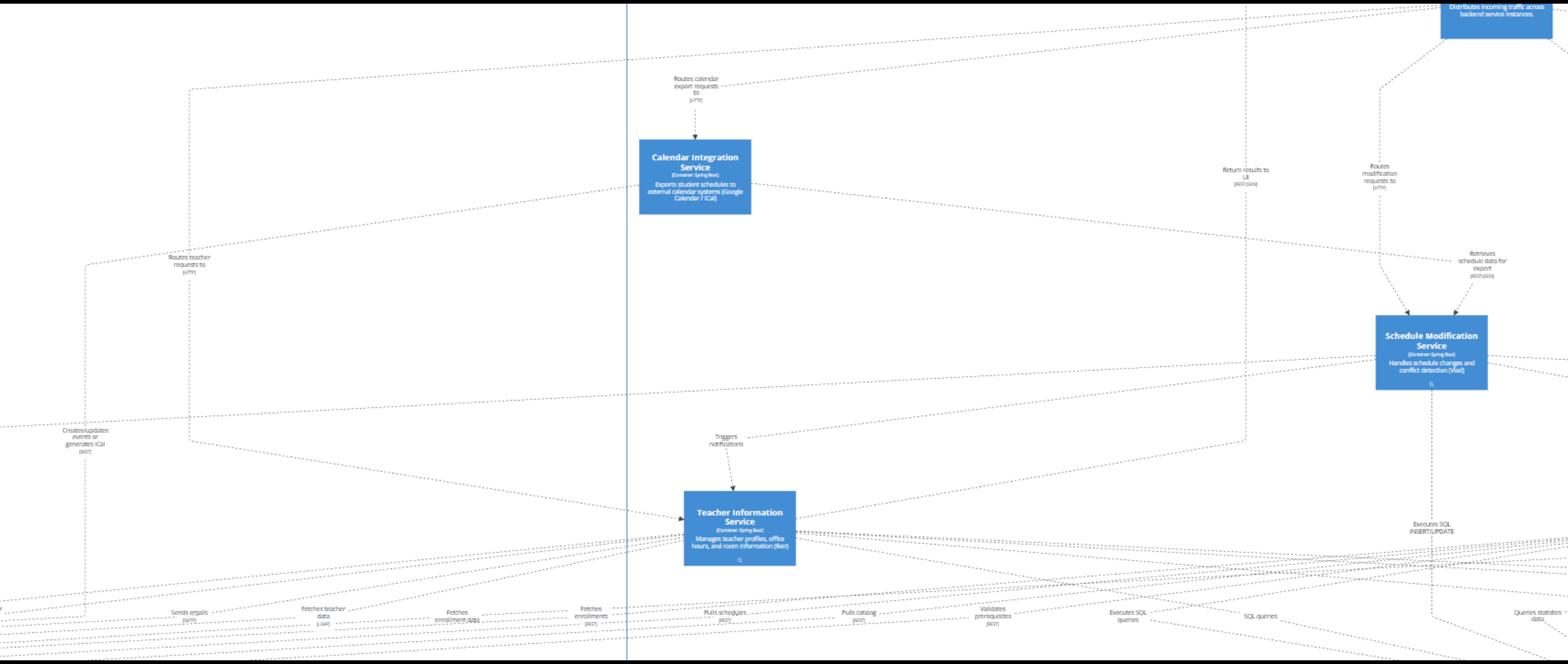
Modifications to the C4 Model

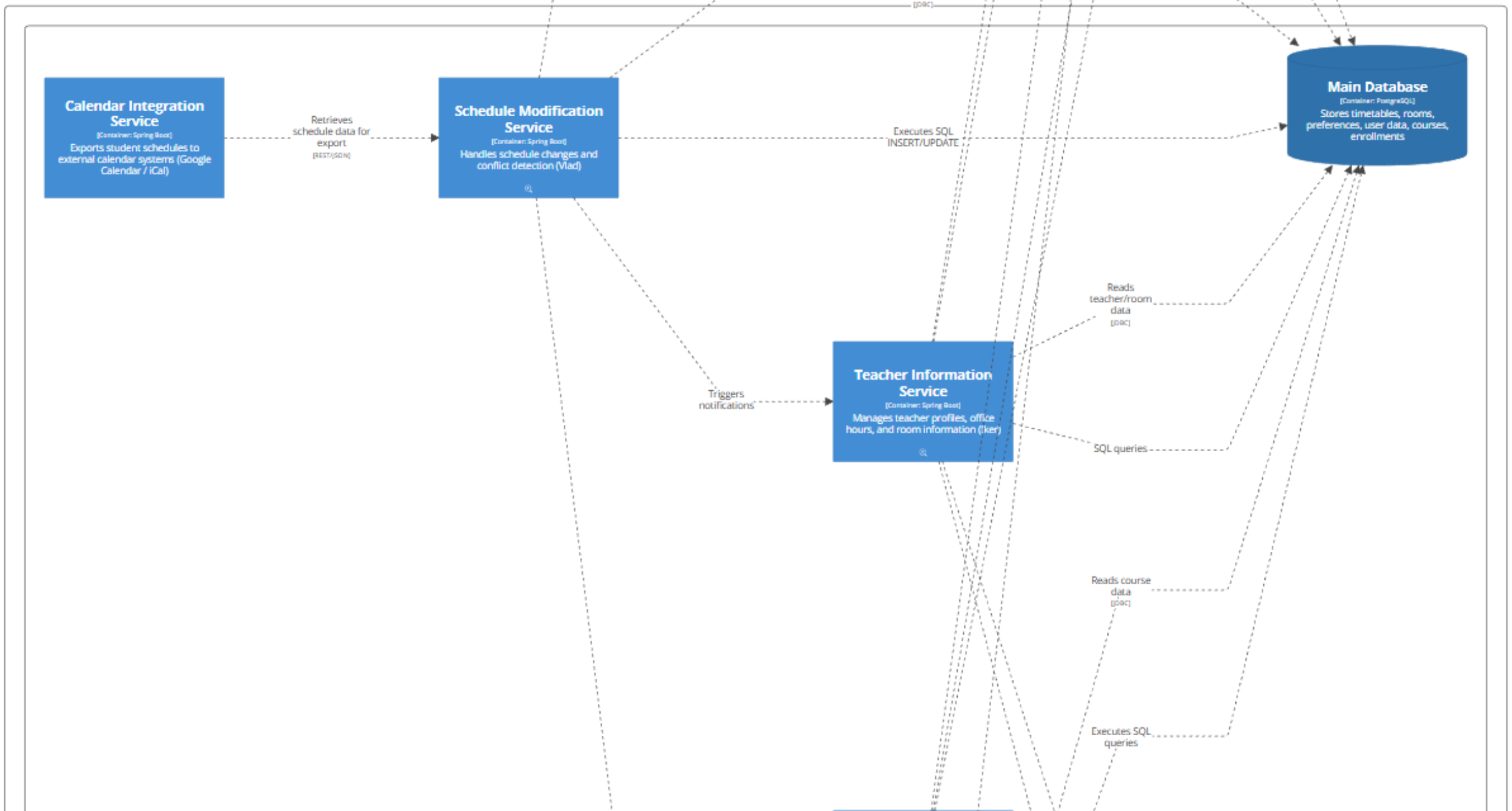
- Container Level:
  A new container CalendarIntegrationService was added to the model. This change is reflected in the Container View.

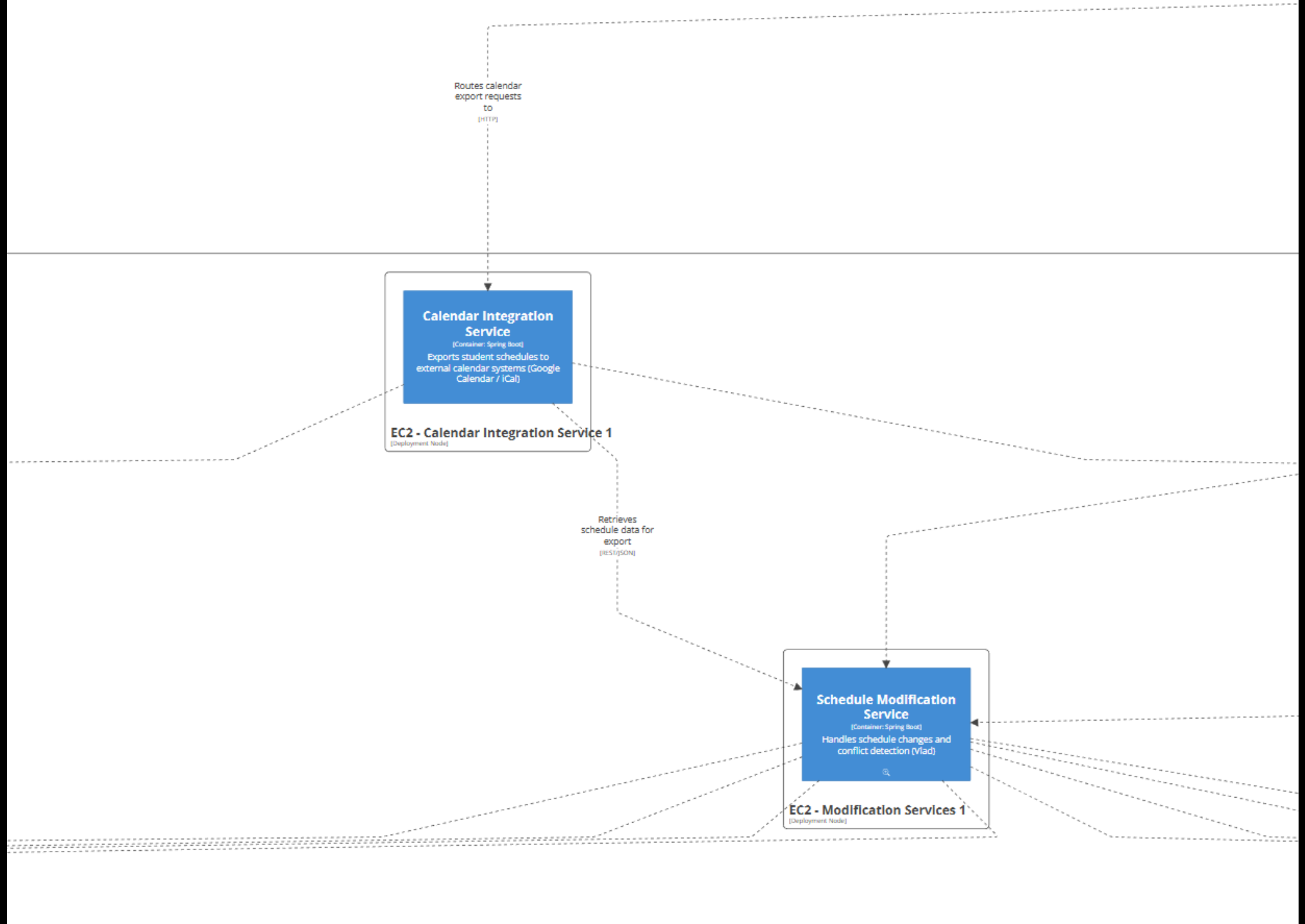
- Relationships:
  New relationships were introduced between the Load Balancer and the Calendar Integration Service, between the Calendar Integration Service and the Schedule Modification Service, and between the Calendar Integration Service and the External Calendar System. These changes are visible in both the System Context View and the Container View.

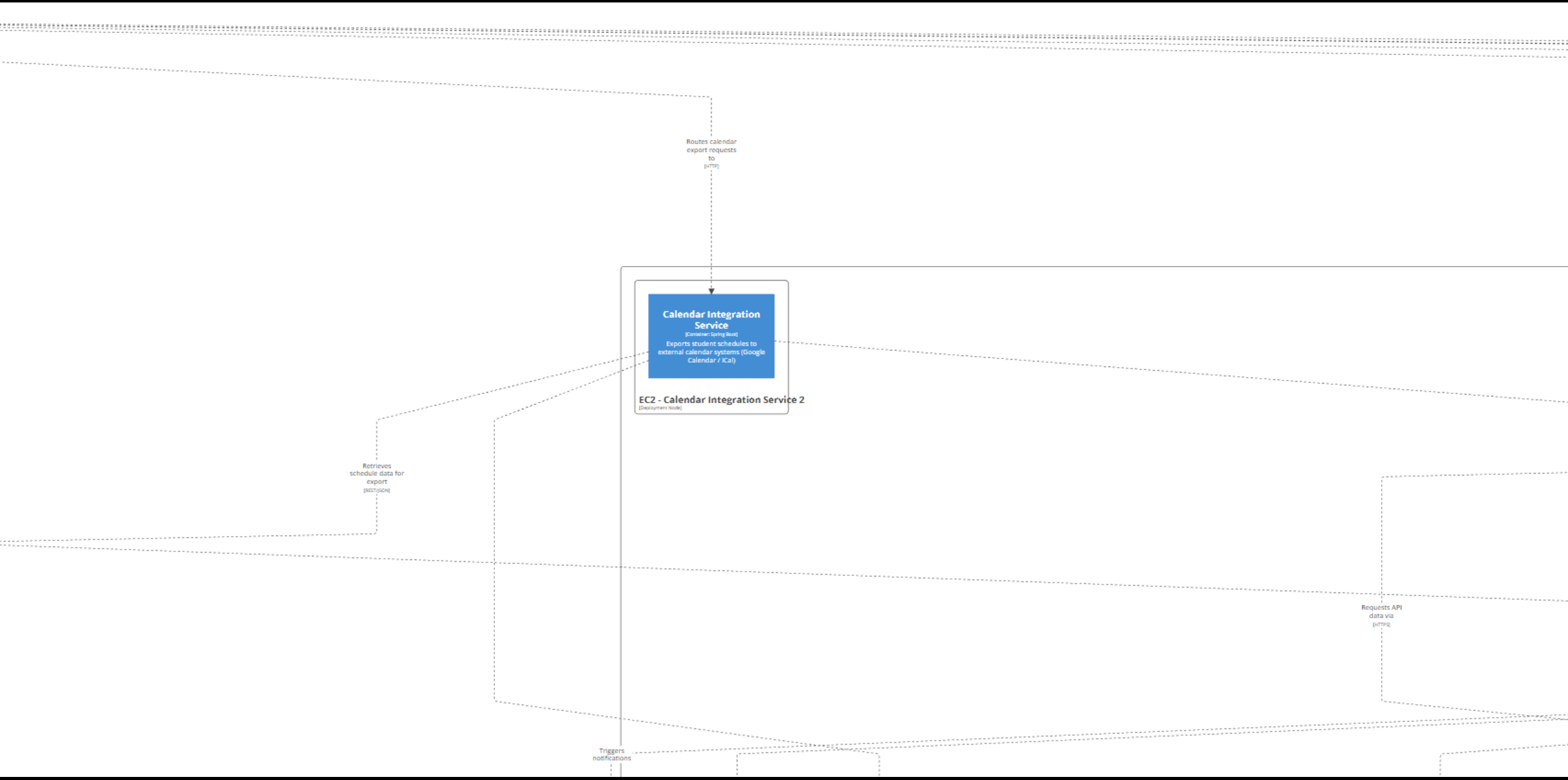
- Deployment:
  The new service was added as a container instance in the Development environment and deployed in two availability zones in the Production environment. These changes are reflected in the Development and Production Deployment Views.

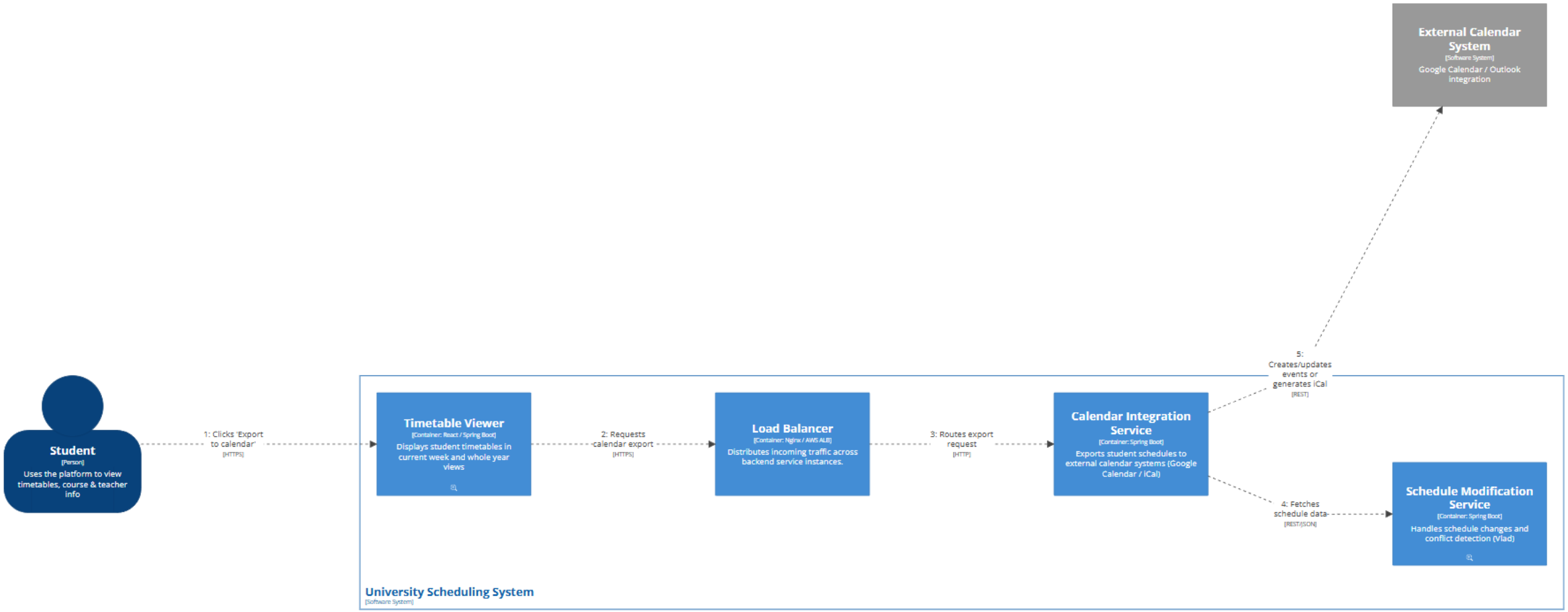
- Dynamic View:
  A new dynamic view was added to document the runtime interaction when a student exports a schedule to an external calendar.


Distributes incoming traffic across backend service instances.
Routes calendar export requests to
[HTTP]
Calendar Integration Service
[Container: Spring Boot]
Exports student schedules to external calendar systems (Google Calendar / iCal)
Return results to UI
[REST/JSON]
Routes modification requests to
[HTTP]
Retrieves schedule data for export
[REST/JSON]
Routes teacher requests to
[HTTP]
Schedule Modification Service
[Container: Spring Boot]
Handles schedule changes and conflict detection (Vlad)
Creates/updates events or generates ICal
[REST]
Triggers notifications
Teacher Information Service
[Container: Spring Boot]
Manages teacher profiles, office hours, and room information (Ilker)
Executes SQL INSERT/UPDATE
Sends emails
[SMTP]
Fetches teacher data
[LDAP]
Fetches enrollment data
Fetches enrollments
[REST]
Pulls schedules
[REST]
Pulls catalog
[REST]
Validates prerequisites
[REST]
Executes SQL queries
SQL queries
Queries statistics data


Updates schedule data
[JDBC]
Calendar Integration Service
[Container: Spring Boot]
Exports student schedules to external calendar systems (Google Calendar / iCal)
Retrieves schedule data for export
[REST/JSON]
Schedule Modification Service
[Container: Spring Boot]
Handles schedule changes and conflict detection (Vlad)
Executes SQL INSERT/UPDATE
Main Database
[Container: PostgreSQL]
Stores timetables, rooms, preferences, user data, courses, enrollments
Reads teacher/room data
[JDBC]
Triggers notifications
Teacher Information Service
[Container: Spring Boot]
Manages teacher profiles, office hours, and room information (Iker)
SQL queries
Reads course data
[JDBC]
Executes SQL queries


Routes calendar export requests to
[HTTP]
Calendar Integration Service
[Container: Spring Boot]
Exports student schedules to external calendar systems (Google Calendar / iCal)
EC2 - Calendar Integration Service 1
[Deployment Node]
Retrieves schedule data for export
[REST/JSON]
Schedule Modification Service
[Container: Spring Boot]
Handles schedule changes and conflict detection (Vlad)
EC2 - Modification Services 1
[Deployment Node]


Routes calendar export requests to
[HTTP]
Calendar Integration Service
[Container: Spring Boot]
Exports student schedules to external calendar systems (Google Calendar / ICal)
EC2 - Calendar Integration Service 2
[Deployment Node]
Retrieves schedule data for export
[REST/JSON]
Requests API data via
[HTTPS]
Triggers notifications



Dynamic View: University Scheduling System

Flow when a student exports a schedule to an external calendar

вторник, 6 января 2026 г. в 11:42 Центральная Европа, стандартное время

Adriana González Nieves

First day of semester, traffic spikes to 5,000 concurrent students accessing "Timetable Viewer"

- **Source of Stimulus:** Student

- **Stimulus:** 5,000 concurrent users attempt to view their timetables simultaneously at 9:00 AM on the first day of the semester.

- **Artifact:** universitySchedulingSystem.timetableViewer, universitySchedulingSystem.loadBalancer (New), and backend services (courseInfoContainer, teacherInfoContainer).

- **Environment:** Production, peak load, high concurrency.

- **Response:** The loadBalancer intercepts the massive influx of requests from the timetableViewer (UI) and distributes them across multiple backend service instances using a round-robin strategy. This prevents any single service instance from becoming a bottleneck and crashing.
- **Measure:** Average page load time remains **under 2 seconds** with **0% error rate** (no timeouts).

Updated diagram